In [2]:
```
import pandas as pd
dataset=pd.read_csv("social_Network.csv")
dataset
```

Out[2]:

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
...	...	...	...	...	...
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

400 rows × 5 columns

In [3]:
```
dataset=pd.get_dummies(dataset,drop_first=True)
dataset
```

Out[3]:

	User ID	Age	EstimatedSalary	Purchased	Gender_Male
0	15624510	19	19000	0	True
1	15810944	35	20000	0	True
2	15668575	26	43000	0	False
3	15603246	27	57000	0	False
4	15804002	19	76000	0	True
...	...	...	...	...	...
395	15691863	46	41000	1	False
396	15706071	51	23000	1	True
397	15654296	50	20000	1	False
398	15755018	36	33000	0	True
399	15594041	49	36000	1	False

400 rows × 5 columns

In [4]:
```
dataset=dataset.drop(columns=["User ID"])
dataset
```

Out[4]:

	Age	EstimatedSalary	Purchased	Gender_Male
0	19	19000	0	True
1	35	20000	0	True
2	26	43000	0	False
3	27	57000	0	False
4	19	76000	0	True
...	...	...	...	...
395	46	41000	1	False
396	51	23000	1	True
397	50	20000	1	False
398	36	33000	0	True
399	49	36000	1	False

400 rows × 4 columns

In [4]:
```
dataset["Purchased"].value_counts()
```

Out[4]:

```
Purchased
0    257
1    143
Name: count, dtype: int64
```

In [5]:
```
x=dataset[["Age","EstimatedSalary","Gender_Male"]]
y=dataset[["Purchased"]]
```

In [9]:
```
dataset.shape
```

Out[9]:

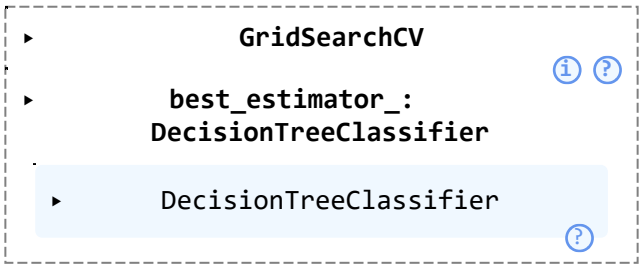
```
(400, 4)
```

In [6]:
```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=1/3,random_state=0)
```

In [7]:
```
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import GridSearchCV
p_g={
    'criterion':['gini','entropy'],
    'splitter':['best', 'random'],
    'max_features':['sqrt','log2'],
    'max_leaf_nodes':[3,6,9,12]
}
clf= GridSearchCV(DecisionTreeClassifier(),p_g,refit=True,verbose=3,n_jobs=-1)
clf.fit(x_train,y_train)
```

```
Fitting 5 folds for each of 32 candidates, totalling 160 fits
```

Out[7]:



In [8]:
```
y_pred=clf.predict(x_test)
y_pred
```

Out[8]:

```
array([0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 1,
       0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0,
       1, 0, 0, 1, 0, 1, 1, 0, 0, 0, 1, 1, 0, 0, 1, 0, 0, 1, 0, 0, 0, 1,
       0, 0, 0, 1, 1, 0, 0, 1, 0, 0, 1, 0, 1, 1, 1, 1, 0, 0, 1, 1, 0, 1,
       1, 0, 0, 1, 0, 0, 0, 0, 0, 1, 1, 0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0,
       0, 0, 0, 1, 1, 1, 1, 0, 1, 1, 0, 1, 1, 0, 1, 0, 1, 1, 1, 0, 0, 1,
       0, 0])
```

In [9]:
```
from sklearn.metrics import confusion_matrix
cm=confusion_matrix(y_pred,y_test)
print(cm)
```

```
[[75 10]
 [10 39]]
```

In [10]:
```
from sklearn.metrics import classification_report
clf_report = classification_report(y_test, y_pred)
print(clf_report)
```

```
              precision    recall  f1-score   support

           0       0.88      0.88      0.88        85
           1       0.80      0.80      0.80        49

    accuracy                           0.85       134
   macro avg       0.84      0.84      0.84       134
weighted avg       0.85      0.85      0.85       134
```

In [ ]: